SPACE INVENTOR

SI-UCAR-WC-RP-004

WindCube

AOCS Design Report

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1 Introduction

This document is produced as part of the WindCube Mission.

The document outlines the AOCS design for the WindCube mission along with an analysis to show compliance with the mission requirements.

The mission from the AOCS perspective can be summarised as guaranteeing that the payload boresight is pointing to the 250 km tangent height to the Earth's surface during observation periods.

The AOCS shall also facilitate the Ground station pointing to optimise the link budget during payload data download.

1.1 Reference Documents

Ref.	Document ID.	Title	Rev.
[RD01]	-	WindCube Requirements Traceability Matrix (RTM), 3.0.0	7 (2024-08-13)
[RD02]	SI-UCAR-WC-RP-001	System Design Report	1.2
[RD03]	WC-SE-0003	WindCube Concept of Operations	3 (2022-01-26)
[RD04]	THRF-NOTE	Tangent Height Reference Frame Note	2024-09-20
[RD05]	Basilisk-dualHingedRigidBodyStateEffector	DUALHINGEDRIGIDBODYSTATEEFFECTOR	1.1

1.2 Abbreviations

BoL	Beginning of Life
CoG	Centre of Gravity
EoL	End of Life
ADCS	Attitude Determination and Control System
RMS	Root Mean Square
APE	Absolute Pointing Error
RPE	Relative Pointing Error
AKE	Absolute Knowledge Error
AGE	Absolute Guidance Error
SIRF	Star Tracker Interface Reference Frame
RTM	Requirement Traceability Matrix
THRF	Tangent Height Reference Frame

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OBC	OnBoard Computer
RAM	Impacting/RAMming direction
RD	Rate Dampening
IAM	Initial Acquisition Mode
SFM	SaFe Mode
CP	Coarse Pointing mode
DEP	DEPloyment mode
NOM	NOMinal mode
FP	Fine Pointing mode
SL	SLew mode
GSP	Ground Station Pointing mode
BSL	Back SLew mode
CAN	Controller Area Network
CSP	Cubesat Space Protocol
SGP4	Simplified perturbations models
TLE	Two Line Elements
MEKF	Multiplicative Extended Kalman filter
LQR	Linear Quadratic Regulator
ACS	Attitude Control System
CAD	Computer Aided Design
BRF	Body Reference Frame

1.3 Scope

The scope of this document is to provide insight into the AOCS solution and expected performance. The document will cover the following topics:

- AOCS Design
  - Driving requirements
  - Overview of the AOCS hardware and architecture
  - Operational modes
  - Reference frames relevant to AOCS
  - Expected interfaces used for Payload data enrichment
- Analysis
  - Assessment of the expected disturbances
  - Solar panel flex mode analysis
- Verification Plan
  - Detailed description of the validation approach
  - Simulation description
- Expected performance
  - Performance assessment for each operational mode

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2 AOCS Design

2.1 Driving Requirements

The key objective of the Windcube mission is to measure the thermospheric winds. The measurements are required to be performed with the payload telescope boresight pointing along track (In the velocity direction of the spacecraft) and cross-track (perpendicular to the orbit plane). The AOCS shall altinate between these two attitude configurations, one orbit at the time.

The payload boresight shall point to the tangent height of 250 km over the Earth's surface.

The driving system requirements for the attitude control system on-board performance are defined in the RTM [RD01] and summed up here as:

- SYS.52 APE < 100 arcsec (RMS) in all axes
- SYS.108 AKE < 30 arcsec (RMS) in all axes
- SYS.109 RPE < 72 arcsec (RMS) over an integration time of 10 sec during payload operations

The performance requirements are all defined relative to the SIRF (Star Tracker Interface Reference Frame).

The agility requirement for Windcube demands a Rest-to-Rest maneuver of 90 deg within 3 minutes including stabilisation to adhere to the APE requirement (SYS.53).

The driving functional requirements for the attitude control system are as defined in the RTM and summed up here:

- WindCube shall track the THRF with shall have an absolute model error less than 5 km relative to the true tangent height
- The AOCS shall have the following five modes:
    - Sun pointing mode
    - Communication mode
    - Cross-track observation mode
    - Along-track observation mode
    - Deorbit mode

2.2 Operational Modes

The AOCS software is executing on the OBC module and performs sensor fusion between all available sensors for coarse and precision attitude knowledge using Extended Kalman filtering techniques, and commands reaction wheels for full 3-axis freedom of the spacecraft to achieve precision pointing for the payload.

Momentum dumping of reaction wheels is performed using 2-axis magnetorquers to maintain a stable momentum of the platform.

The AOCS software is designed to control the attitude of the spacecraft based on the specification of two target vectors and two source vectors:

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- The target vectors specify the direction in which (e.g. sun, nadir, velocity vector) the spacecraft shall be oriented.
- The source vectors specify which surface - or more precisely which vector in the spacecraft body frame coordinate system - that shall be oriented in the directions specified by the target vectors.

The AOCS aligns the primary source vector in accordance with the specified primary target vector. As this constrains only two degrees of freedom, the secondary set of vectors is needed to achieve 3-axis control. Therefore, the AOCS also aligns the secondary source vector as well as possible in accordance with the secondary target vector.

Thus, if the various spacecraft attitude modes to be used throughout the mission are specified in terms of such target and source vectors, the transition between these modes can be commanded simply by changing the values of the four vectors.

The following overall AOCS modes are requested for WindCube:

- **Sun Pointing (SA):** Maximum Power Generation Mode. AOCS shall point solar panels towards the sun while all non-essential systems are shut off.
- **Communications Mode (GSP):** AOCS shall orient for ground station pointing for downlinking
- **Cross-Track Observing (FP):** AOCS shall orient for cross-track observations.
- **Anti-Track Observing (FP):** AOCS shall orient for anti-track observations.
- **Deorbit Mode:** AOCS shall the -Z-axis in the RAM direction to maximise drag

Enabling the AOCS is controlled via parameters defined in the Parameter System. This implies that the enabling can be performed manually via ground command or automatically via a Command Schedule that is started when the spacecraft is powered on. Waiting the required 45 min after separation from the dispenser pod to allow manoeuvring is therefore possible.

Initial Acquisition and Safe mode (IAM/SFM)

The ADCS will enter the Initial acquisition mode right after separation from the launcher. The mode will make sure to dampen the initial tip-off rate in RD before it moves to Sun Acquisition (SA) using the stowed sun pointing configuration.

Once sun is acquired, it will move into the Steady State mode (SS) where it will wait for TeleCommands to move to a higher state of operation (Nominal mode) or enter the Deployment mode (DEP). In deployment mode, active attitude control will be disabled, until deployment of the solar panels has been confirmed and the command to return to Rate Damping (RD) has been received. This is to guarantee that the motion excited from the solar panel deployment does not feed back into the controller, creating unwanted oscillations.

The difference between IAM and SFM is merely to highlight the difference between the Sun Acquisition pointing axis in Stowed and deployed configuration. IAM is only used once in the early stage of the mission.

Nominal Mode (NOM)

Once Nominal mode is entered, 3-axis attitude knowledge is acquired using gyro, magnetometer and Fine Sun Sensors. Once the onboard attitude estimator has converged, the ADCS will automatically move to Coarse pointing mode (CP). While in CP the, the ADCS will track primary to Nadir and Secondary to Sun.

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Once in Coarse Pointing (CP), the ADCS is ready to enter the mission mode also called Fine Pointing (FP) which will utilise the Star Tracker for increased accuracy on attitude knowledge. In this submode, the ADCS will track the Tangent height Reference frame in either Along- or Cross-Track configuration.

The ADCS will fallback to CP if the Star Tracker is not available.

While in CP or FP, a change in attitude reference will switch mode to Slew (SL) and return to CP or FP once the pointing error has converged.

From CP and FP, ground station pointing (GSP) can be commanded or configured to automatic transition. This will trigger a switch to SL →GSP →BSL →CP/FP. The back slew (BSL) will make sure to return to the original attitude reference, before entering the SL.

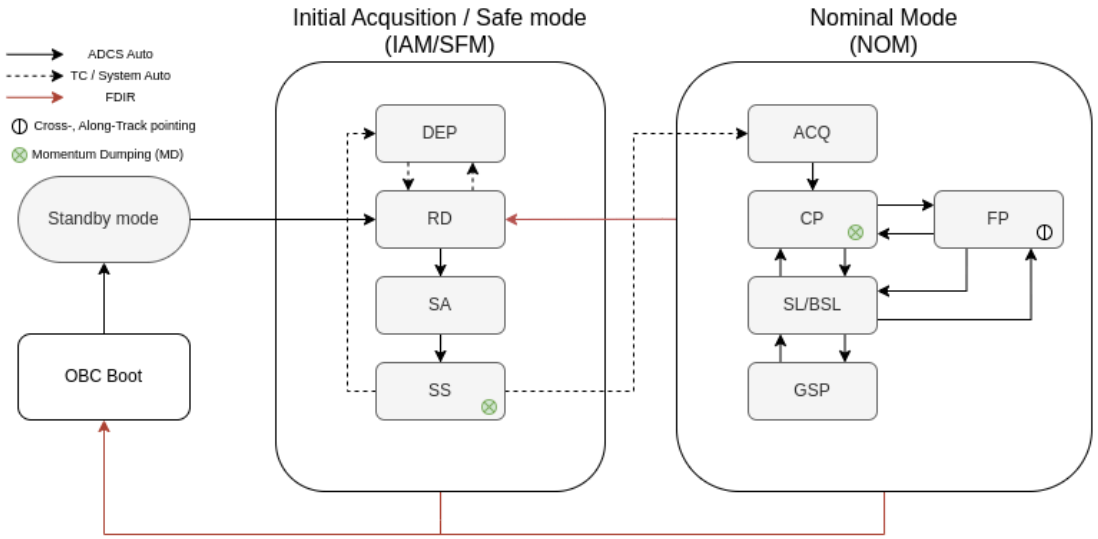

Figure 2-1: The ADCS operational modes and submodes

Equipment	IAM/SFM	NOM
MAG	MD	ACQ/MD
FSS	SA/SS	ACQ/CP
GYRO	RD/SA/SS	ACQ/CP
STR	-	FP/GSP
GNSS	-	CP/FP/GSP
MQT	RD/MD	MD
RW	SA/SS	CP/FP

Table 2-1: Shows the how the equipment is used in the different modes and submodes

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2.3 Equipment and Accommodation

The following figures and table shows the Actuators and Sensors accommodated by the ADCS and connected on the CAN bus and communication via CSP.

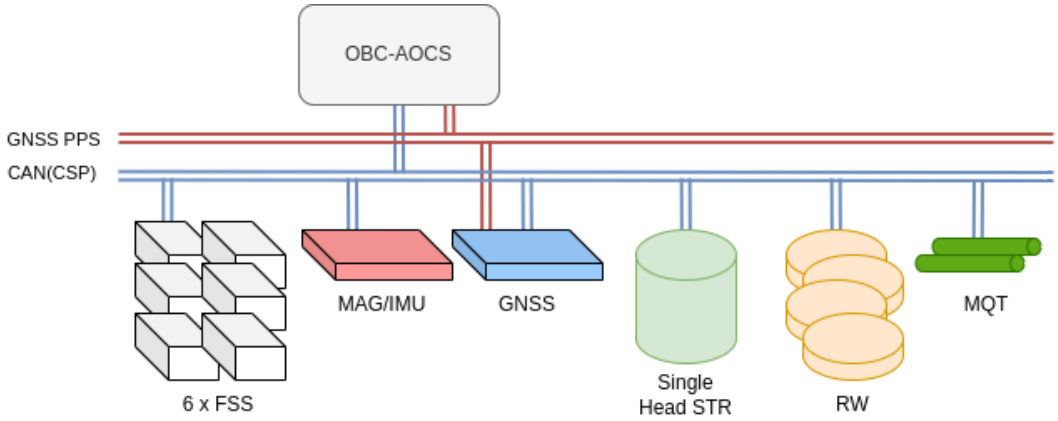

Figure 2-3: The ADCS related equipment powered through the on-board PDU, and connected via the redundant CAN interface over CSP.

Unit	Pcs	Type	Unit name
OBC	1	ARM Cortex M7 processor - Including a cold redundant core as fallback ADCS	OBC-P4
IMU/MAG	1	Integrated inertial measurement unit with Gyro and Magnetometer (A single unit includes a redundant sensor pair)	IMU-P4
FSS	6	3-axis Fine Sun Sensor	FSS-P4
GNSS	1	Guidance Navigation Satellite System w. PPS	GNSS-P4
STR	1	Optical Star tracker head w. Rate estimate	Jena Optronik - ASTRO-CL
MQT	2	2 Am^2 Single axis Magnetorquers	MQT-200-P4
RW	4	100 mNms Reaction Wheels	RW-100-P4

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2.4 AOCS Architecture

2.4.1 ADCS Algorithm

The AOCS software is built around Space Inventor's software stack, using CSP and the parameter system to receive commands and provide telemetry.

The main features of the ADCS software is a modular and configurable design:

- Orbit Estimators:
  - TLE based SGP4/SDP4 propagator or GNSS
- Flexible Reference System:
  - Primary and secondary target and source vectors allow accurate primary pointing with the one remaining DoF optimising secondary pointing
- Tunable Attitude Determination:
  - MEKF, tunable via. the parameter interface.
- Tunable Attitude Control:
  - LQR based attitude control design. Tunable via. the parameter interface.
- Implementation for common sensor types
- An ephemeris implementation to estimate the sun, moon and earth position. The sun position is used eg. with sun sensors and for Beta-tracking.
- Updatable magnetic field model.
- Model parameters are exposed to param, and can be changed when new parameters are released.
- Reaction Wheel Bias Controller
  - Bias the reaction wheels to some reference momentum. Utilises the null-space of the wheel cluster along with magnetorquers.

The fundamental structure of the algorithm and its relationship to the environment can be depicted by Figure 2-4, excluding the control loop for momentum management and dumping.

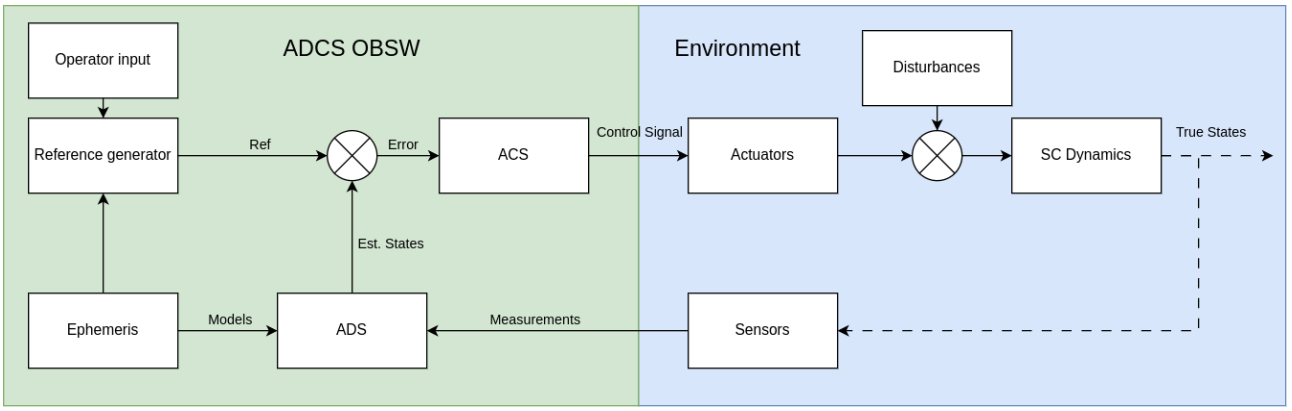

Figure 2-4 Shows the general structure of the ADCS algorithms

ADS - Attitude Determination System will use the available sensor information and perform sensor fusion by comparing with predicted measurement using a Multiplicative Kalman Filter. The Kalman filter is running with Gyro in dynamic replacement mode and estimating the attitude quaternion and gyro bias.

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ACS - The Attitude Determination System is based on a 3-axis LQR for attitude stabilisation and fine steering. Based on a reference state and the estimated state vector, the controller calculates a control signal for the reaction wheels which will force the difference to zero and stabilise the satellite.

Ephemeris - Is a collection of models which predicts the satellites position and velocity. Moreover it also provides a prediction of the Sun position and the Earth magnetic field strength and direction. This information is used for Attitude estimation and for attitude reference generation.

Reference generator - Calculates the 3-axis attitude and angular velocity reference, based on configuration, operator input and Ephemeris.

Sensors - The true attitude will be perceived through the avionics sensors indirectly and estimated through sensor fusion.

Actuators - The control signal from the ACS is converted into the coordinate system of the individual actuator, where the actuator calculates the necessary control signal to accelerate the wheels, providing the commanded torque.

Disturbances - Is the collective external disturbance torques acting on the satellite body. These are in general:

- Aerodynamic pressure torques
- Internal magnetic moment
- Gravity gradient torques
- Solar radiation pressure

These disturbance torques will be compensated for by the ACS using the closed-loop controller. The torque will accumulate in the reaction wheels over time and will have to be unloaded using the magnetorquers.

SC Dynamics - The sum of all external and internal torques will accelerate the satellite angular velocity and change the attitude of the spacecraft.

2.4.2 Reference frames

2.4.2.1 Satellite Body Reference Frame (BRF)

The BRF is as defined from the CAD:

-x : Payload boresight

+y : Star Tracker boresight

+z : Completed the right-handed euclidean basis

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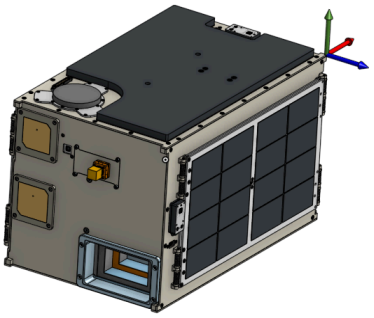

Figure 2-5: Shows the windcube body reference frame (BRF) in stowed configuration

Windcube will be deployed in the stowed configuration where the ADCS will enter the Initial acquisition mode (IAM) and go to Sun pointing. Here the Sun pointing direction will be configured to point the -z-axis to the Sun.

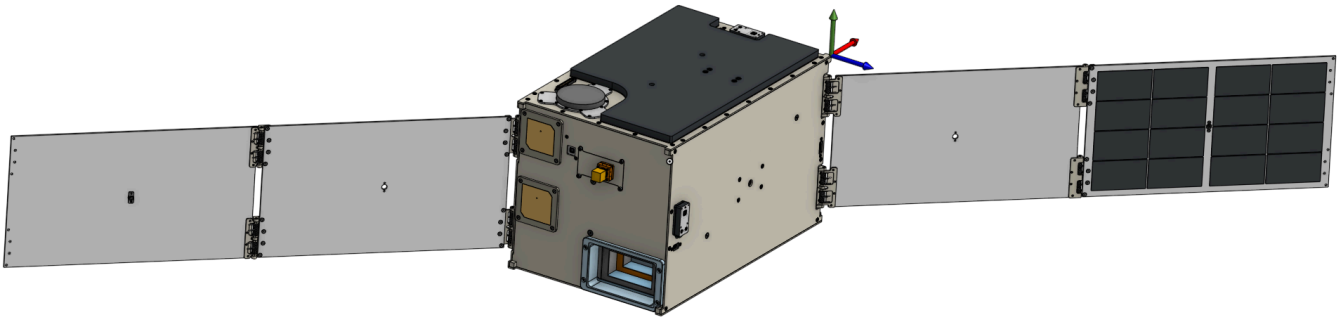

Figure 2-6: Shows the Windcube body reference frame (BRF) in deployed configuration

Once deployed, the ADCS will enter into Safe Mode (SFM) and be reconfigured to point the +x/-z 45 deg vector to Sun. This will maximise the power input.

2.4.2.2 Tangent Height Reference Frame (THRF)

For the WindCube mission, anti- and cross-track observations shall be performed, which will require an additional reference frame to be implemented. This will enable the AOCS to track the Tangent height of 250km.

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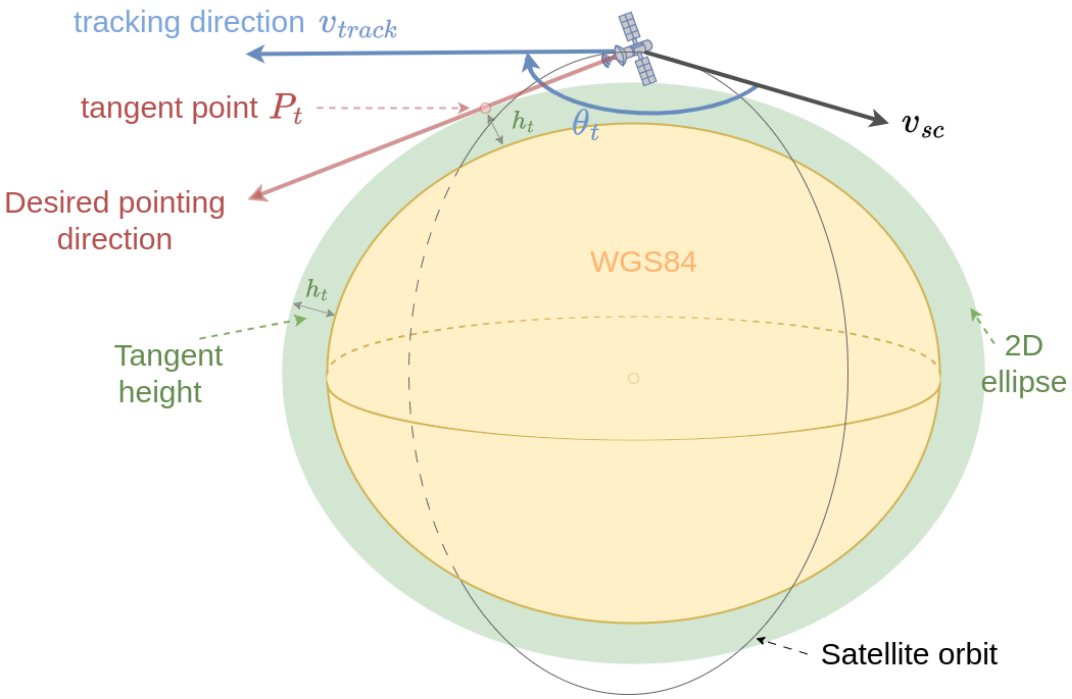

Figure 2-7: Definition of Tangent Height Reference Frame

The Tangent Height Reference Frame (THRF) is the reference frame used during payload operations. Its purpose is to follow a point above the Earth surface of 250 km +/- 5 km, tangent of the boresight look direction to the 250km airglow region [RD03].

Effectively, this point moves with the satellite orbit, in the direction of satellite velocity. The reference frame will be defined relative to these criterias.

The THRF shall have two configurations, which are anti- and cross-track observations. For anti-track observations, the tangent height point is located in the orbit plan as depicted in Figure 2-7. For cross-track observations, the tangent height is in the plane which is perpendicular to the orbit plane.

The theory behind the THRF and validation of it is documented in the reference document [RD4].

In the following figures, the satellite BRF is shown relative to the Tangent height reference frame to illustrate the two configurations Along- and Cross-Track.

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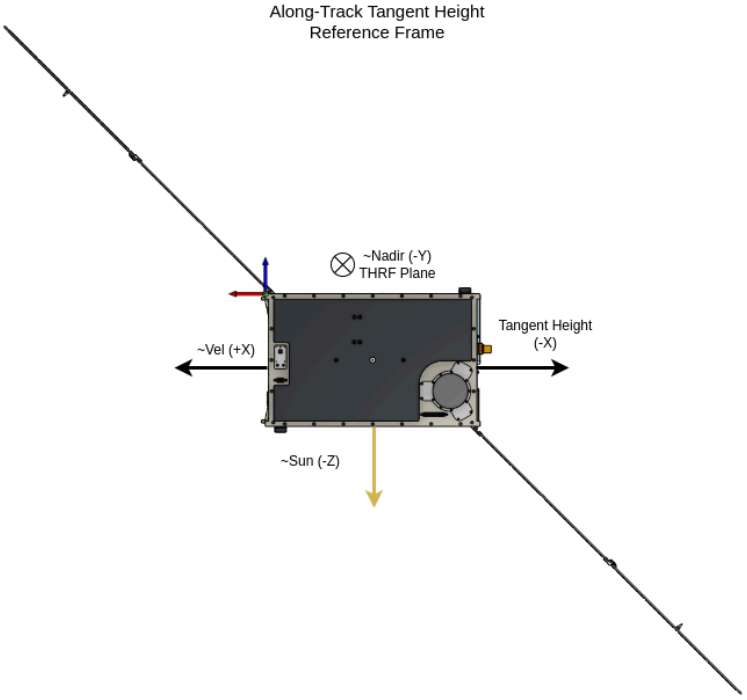

Figure 2-8: THRF - Along-track configuration relative to satellite BRF. The tilde in front of the axis symbolises an approximate direction.

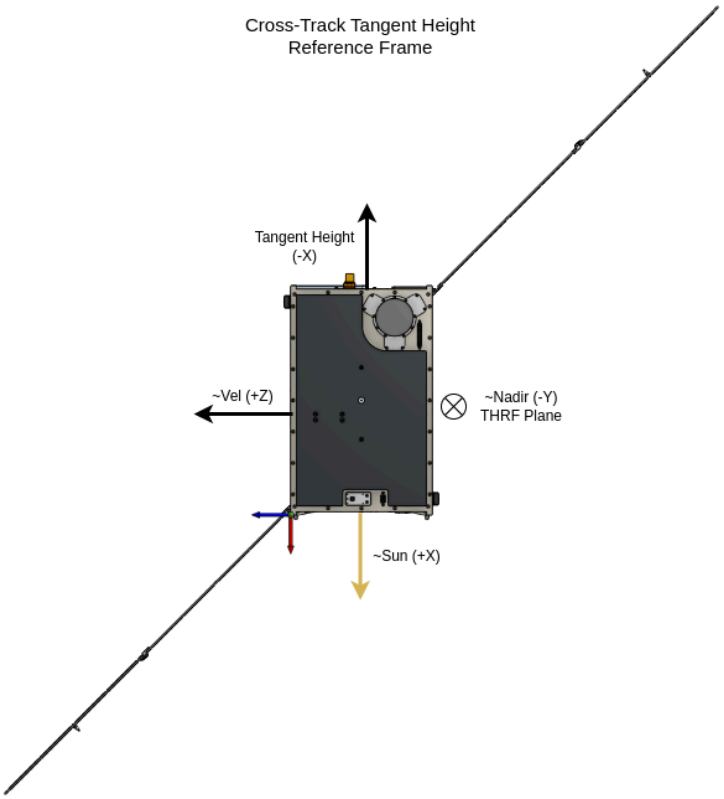

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Figure 2-9: THRF - Cross-track configuration relative to satellite BRF. The tilde in front of the axis symbolises an approximate direction.

2.4.2.3 Star Tracker Interface Reference Frame (SIRF)

The Star Tracker Interface Reference Frame (SIRF) Coordinate system definition, is where +z is normal to the plane of the aluminium lid, pointing in the direction of the FOV, +x is defined as parallel to the edge of the lid. +y completes the right-handed coordinate system

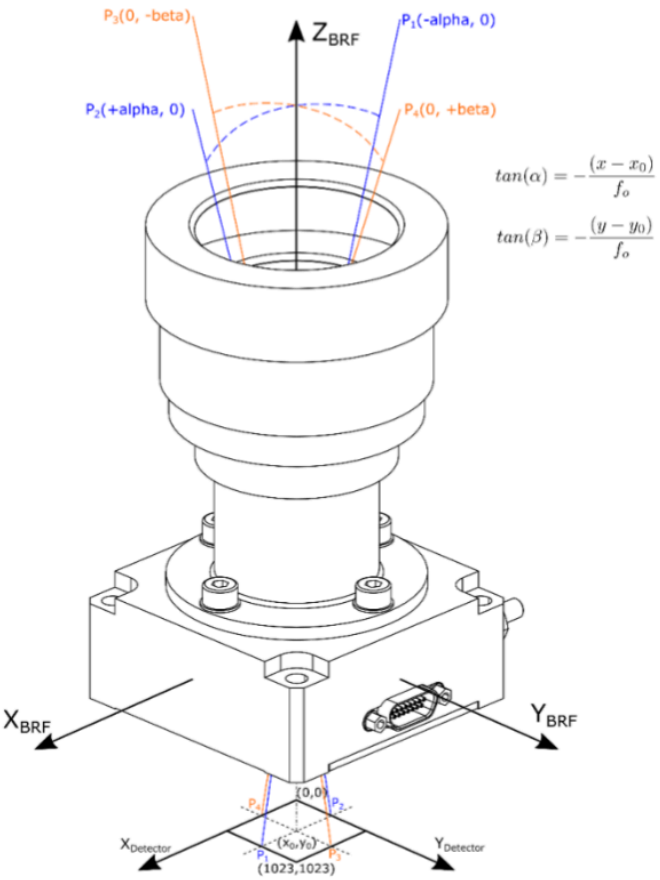

Figure 2-10: Definition of the Star tracker Interface Reference Frame (SIRF). Supplier documentation is using the BRF notation.

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3 Analysis

The following Franck diagram illustrates the chain of coordinate system and transformation errors used for pointing assessment of WindCube. The coordinate frames are described hereafter.

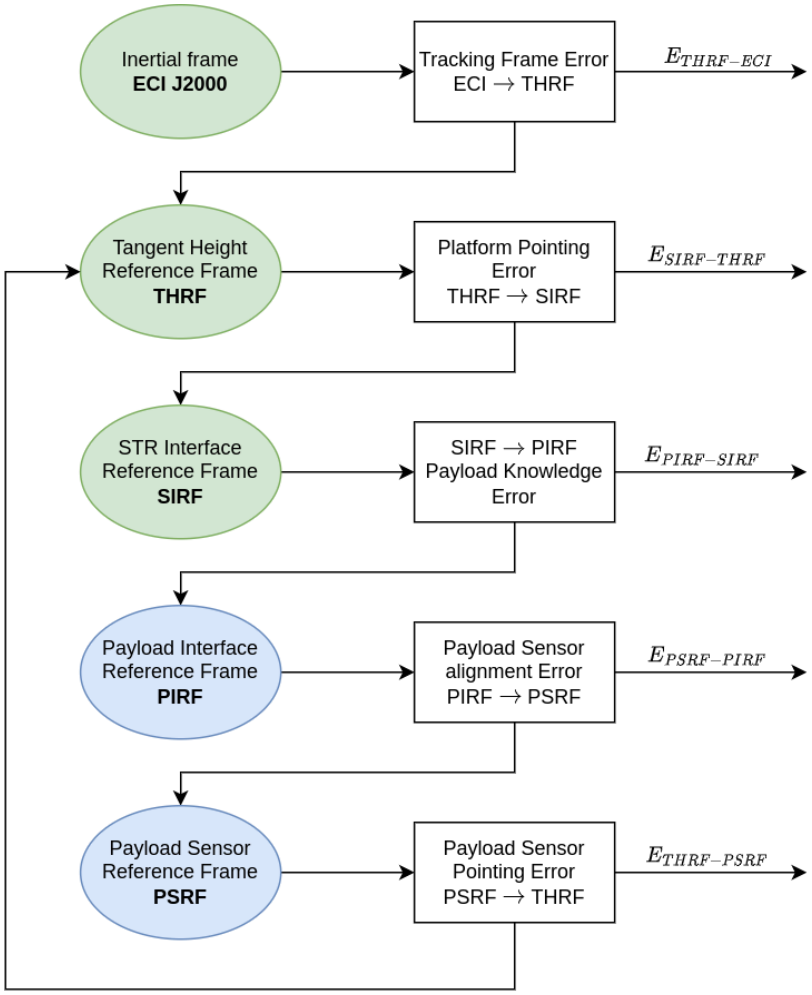

Figure 3-1: Franck diagram with WindCube reference frames

Table 3-1: Pointing assessment

Error Group	Pointing Assessment
$E_{THRF-ECI}$	The tracking frame error represents the model error which occurs from deriving the reference frame based on: • WGS84 model errors • Satellite PV uncertainty • Time delay
$E_{SIRF-THRF}$	Representing the platform pointing error, which includes all the error contributions from the list below. These pointing contributors will be investigated and mitigated as part of the ongoing analysis.

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<table>
<tr><td rowspan="3"></td><td colspan="2">Moreover, each contributor will have a direct or indirect impact on the APE, AKE and RPE.</td></tr>
<tr><td>Time-constant errors</td><td>Time-varying errors</td></tr>
<tr><td>• Star tracker
  ○ Bias
• Reaction wheel misalignment</td><td>• Star tracker
  ○ FOV Error
  ○ Pixel error
  ○ Temporal Error

• Reaction wheel
  ○ Torque noise
  ○ Micro-vibration

• Environmental disturbances
  ○ Aerodynamic drag torque
  ○ Magnetic dipole
  ○ Solar radiation Pressure
  ○ Gravity Gradient

• Satellite Flexible modes
  ○ Solar panel oscillations

• Attitude control error</td></tr>
<tr><td rowspan="3">$E_{PIRF-SIRF}$</td><td colspan="2">Represents the alignment error between the Star Tracker interface SIRF and the Payload Interface PIRF.

This error will have the following contributors, which are <u>not</u> included in the pointing budget, given by the APE requirement [RD01]:</td></tr>
<tr><td>Time-constant errors</td><td>Time-varying errors</td></tr>
<tr><td>• Axial tolerances from SIRF to PIRF</td><td>• Axial thermal deformation between SIRF and PIRF</td></tr>
<tr><td>$E_{PSRF-PIRF}$</td><td colspan="2">Internal alignment error between the Payload Interface Reference Frame PIRF and the Payload Sensor Reference Frame. This alignment error will alongside the $E_{PIRF-SIRF}$ result in the overall payload alignment error.</td></tr>
<tr><td>$E_{THRF-PSRF}$</td><td colspan="2">Represents the payload sensor pointing error with respect to the Tangent Height Reference Frame.</td></tr>
</table>

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3.1 External Disturbances

The Windcube mission will operate in a dusk-dawn orbit at an altitude in the range from 600-500 km. For the disturbance analysis, the lowest altitude is considered. In this orbit, the following disturbances are taken into account:

- Magnetic dipole
- Aerodynamic drag
- Solar radiation pressure
- Gravity gradient

The magnetic dipole is considered small and in the range of 50 mAm2. The satellite CoG displacement relative to the geometric centre is considered less than 30 mm.

The aerodynamic drag and solar radiation pressure torques have been simulated using a geometric model with a mesh of facets.

A day in life while in along-track pointing has been simulated to analyse the accumulated momentum and disturbance magnitudes.

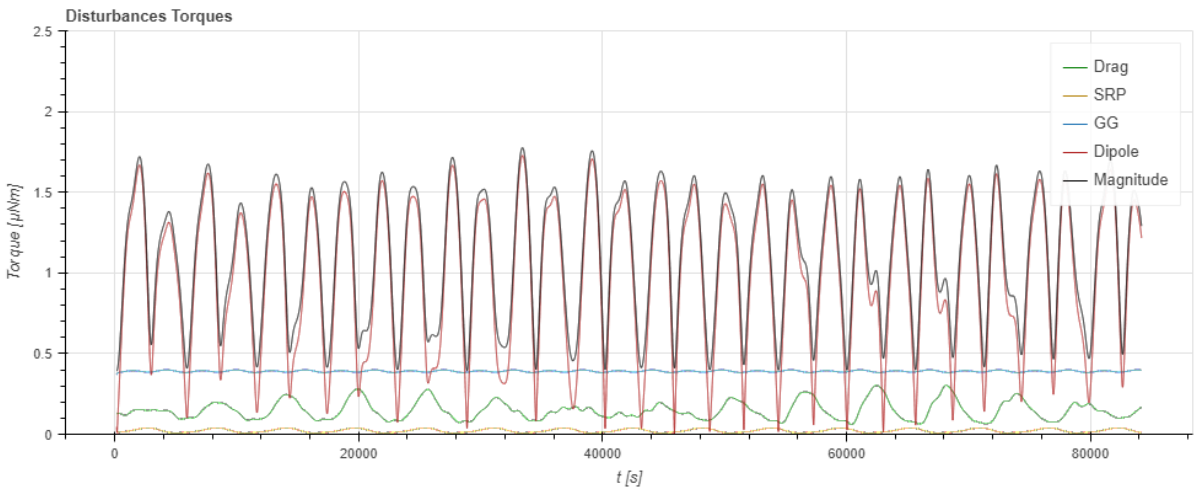

Figure 3-3: Shows the simulated disturbances acting on the satellite over 15 orbits

It can clearly be seen that the magnetic dipole has the largest effect, as expected. The peak disturbance torque lies around 1.7 μNm.

If the ADCS runs with a control loop frequency of 1 Hz then the pointing error contribution from the external disturbance can be found from the double integral of the acceleration.

$$\theta(t) = \frac{1}{2}\frac{\tau_d}{J_{minor}}t^2$$

With J_{minor} baselined to 0.22 kg.m2. Then the question is when the error can be detected and corrected for. The measurement noise of the star tracker is less than 6 arcsec 3-sigma. This means that the disturbance torque can potentially build up over 3 seconds before it can be observed.

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A different approach is to look at the margin for peak disturbance torque. The RPE requirement of 72 arcsec over 10 seconds window can be seen as a limit on the relative angular rate of 7.2 arcsec/s. Based on this knowledge, the peak disturbance torque shall not be greater than 4 μNm or a magnetic dipole moment higher than 88 mAm2. This is the baseline magnetic dipole moment used further in this analysis.

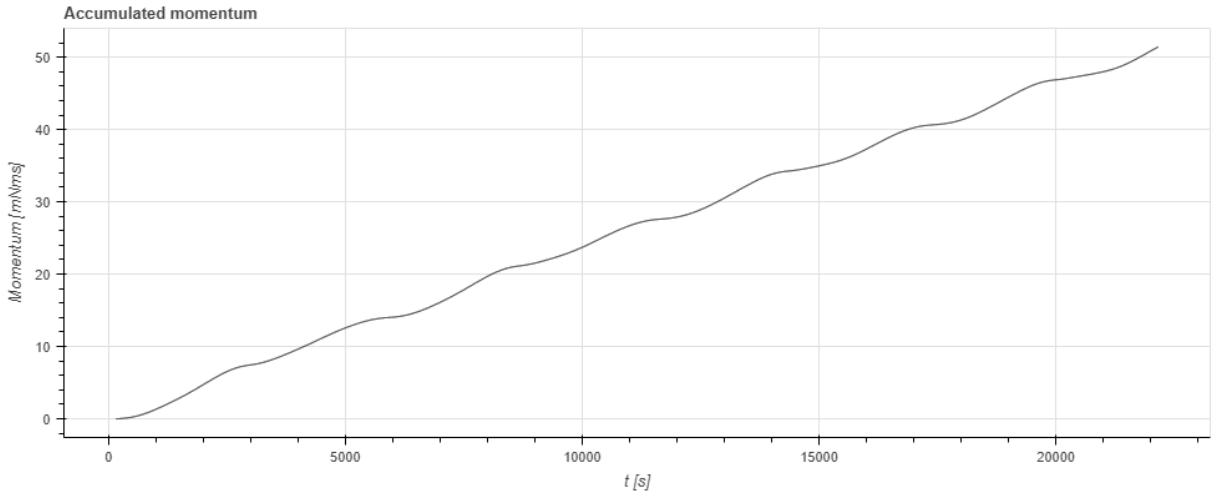

Figure 3-4: Shows the momentum buildup over 4 orbits

Based on the disturbance torques, the momentum will build up in the reaction wheels. For the windcube mission, it's important that the momentum buildup can be stored in the wheels until outside the observation windows. It is clear from the simulation that this is acceptable since one momentum wheel can store 100 mNms.

3.2 Flex mode analysis

Satellite flexible modes are a general concern when investigating attitude stability and controller bandwidth. As per design, Windcube has two dual hinged solar panels which are deployed after separation from the launcher. The solar panels themselves are made from a PCB material, which are by nature not stiff enough to be disregarded in this analysis. The solar panel flexible mode has been analysed and modelled as part of the satellite system.

Cantilever eigenfrequency estimation

For estimating the cantilever eigenfrequencies, an analytical approach has been taken and hereafter cross-checked using a Finite Element Model of the solar panel structure.

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Simple analytical estimation of eigenfrequency

BE beam model

Parameters

$l := 0,644\ \mathrm{m}$	Length
$b := 207,3\ \mathrm{mm}$	Width
$t := 0,002\ \mathrm{m}$	Thickness
$\rho := 1600\ \frac{\mathrm{kg}}{\mathrm{m}^3}$	Density
$E := 24,8\ \mathrm{GPa}$	Youngs modulus
$I := \frac{1}{12} \cdot b \cdot t^3$	MoI

$I = 138,2\ \mathrm{mm}^4$

Formula for eigenfrquency:

$$\omega(x) := \frac{(x)^2}{2 \cdot \pi} \cdot \sqrt{\frac{E \cdot I}{(\rho \cdot (b \cdot t)) \cdot l^4}}$$

here 'x' is the root 'βl', and division by 2π to get [Hz]

First f_n:	Second f_n:	Third f_n:	Fourth f_n:
$\omega(1,875) = 3,0666\ \mathrm{Hz}$	$\omega(4,694) = 19,2194\ \mathrm{Hz}$	$\omega(7,855) = 53,8203\ \mathrm{Hz}$	$\omega(10,99) = 105,3535\ \mathrm{Hz}$

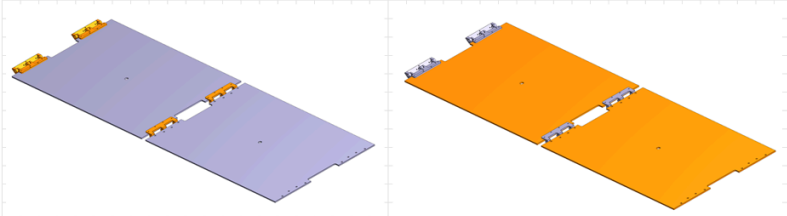
Thus, the beam model actually captures the pure bending mode rather well. The modes that involve torsion are missed by the beam model. In case the hinge material is changed to steel the eigenfrequencies get even closer.

Figure 3-5: Notes from analytical estimation of the cantilever eigenfrequencies.

Eigenfrequencies for solar panels calculated as beams and plates:

Reference FE model

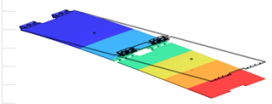
A reference FE model is generated with hinge made from aluminium 6061 and panels of Arlon 85N properties.



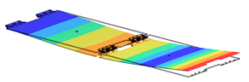
The root hinges are fixed and all points in the hinges are bonded (ie. the hinges does not have any kinematic behaviour, only their material stiffness.

Solving the model gives the following eigenfrequencies:

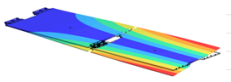
2,85Hz



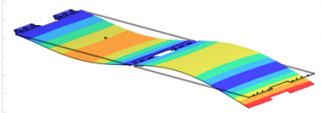
17.8Hz



20.5 Hz



50.8 Hz



62.9 Hz

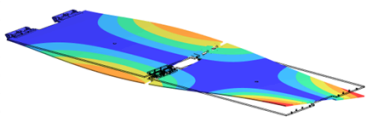


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Figure 3-6: Notes from FEM analysis, validating the analytical approach.

Based on the results from the FEM analysis, it could be concluded that the lowest cantilever eigenfrequency is approximately 2.8 Hz.

The damping of this resonance frequency is unknown and will have to be experimentally validated. From the ADCS analysis point of view, a sub-system requirement will have to be identified, to guarantee enough damping of this eigenfrequency. This will be addressed later in this section.

Satellite system eigenfrequency

Till this point, the eigenfrequencies for a single panel have been estimated, but the total system contains a satellite hub and two dual hinge solar panels. To model the effects of the flexible modes on the satellite system a Newtonian mechanics model has been used.

The model used is documented in [RD5] and developed by Autonomous Vehicle Simulation (AVS) Laboratory and simulated in the Open Source spacecraft dynamics framework called Basilisk.

The model parameters are matched with WindCubes physical Mass, Centre of Mass and Inertias (MCI) for the satellite hub and the solar panels elements. The spring constants and damping coefficients are matched to fit with the cantilever eigenfrequencies found in the FEM analysis of 2.8 Hz.

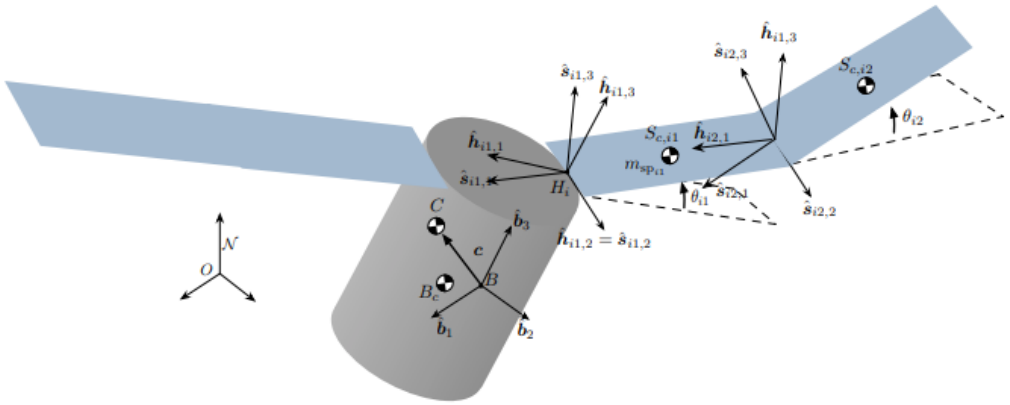

Figure 3-6: Shows the dual hinge flexible spacecraft model [RD5]

System identification has be utilised to find the linear relationships between the the input torque $\tau(s)$, output satellite attitude $\theta(s)$ and angular velocity $\omega(s)$ respectively.

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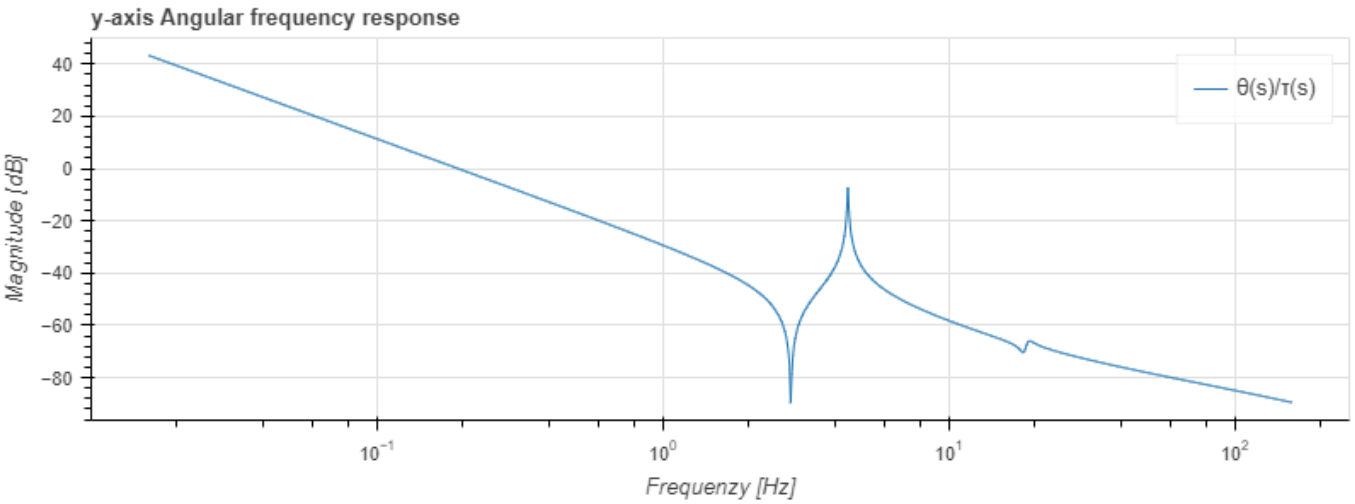

Figure 3-7: Shows the frequency response on satellite angle wrt. Input torque.

From Figure 3-7 it can be seen that the structural eigenfrequency σ_s lies at around 4.45 Hz.

The phase of the open-loop transfer function between $\theta(s)/\omega(s)$ must be less than -180° in the vicinity of crossover frequency ω_{co} region, where the magnitude is 0 dB, to provide the necessary phase margin and stability. The dampening of the structural mode is chosen to achieve an open-loop gain margin of 6 dB resulting in a crossover frequency ω_{co} at 0.2 Hz. This will be the maximum achievable Open-loop bandwidth of the attitude control system.

From a theoretical point of view, we can use this knowledge to backtrace the required critical damping in the flexible solar panels to guarantee stability.

The desired damping ratio ξ in the solar panels shall be higher than 2.6e-3 which translates to a decay rate λ of 0.072 or a half-time of approximately 10 seconds. This shall be validated through practical modal tests of the solar panels. If the damping ratio in the panels is lower than the required, the maximum achievable controller bandwidth will be lowered to achieve a 6 dB gain margin.

Given that this criteria is fulfilled, and the controller bandwidth is less than 0.2 Hz, the satellite structural dynamics can be seen as rigid and stability is guaranteed.

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4 Verification Plan

The ADCS development and validation strategy will follow the V-cycle approach with inspiration from GMV's DDVV approach [RD2]. An incremental DDVV approach based on the following chain will be used:

FES/AIL⇨CIL⇨HIL

Here, FES is meant as a Functional Engineering Simulator which will be used to test the ADCS software in the AIL, CIL and HIL configurations respectively.

The FES is a numerical simulator including a full environment containing the satellite dynamics, sensors, actuators models and disturbances acting on the spacecraft. The FES is able to run faster than real-time and as a RT environment.

The Algorithms will then be tested and validated in an incremental Verification and Validation loop until requirements are satisfied. This is illustrated in Figure 3.1.

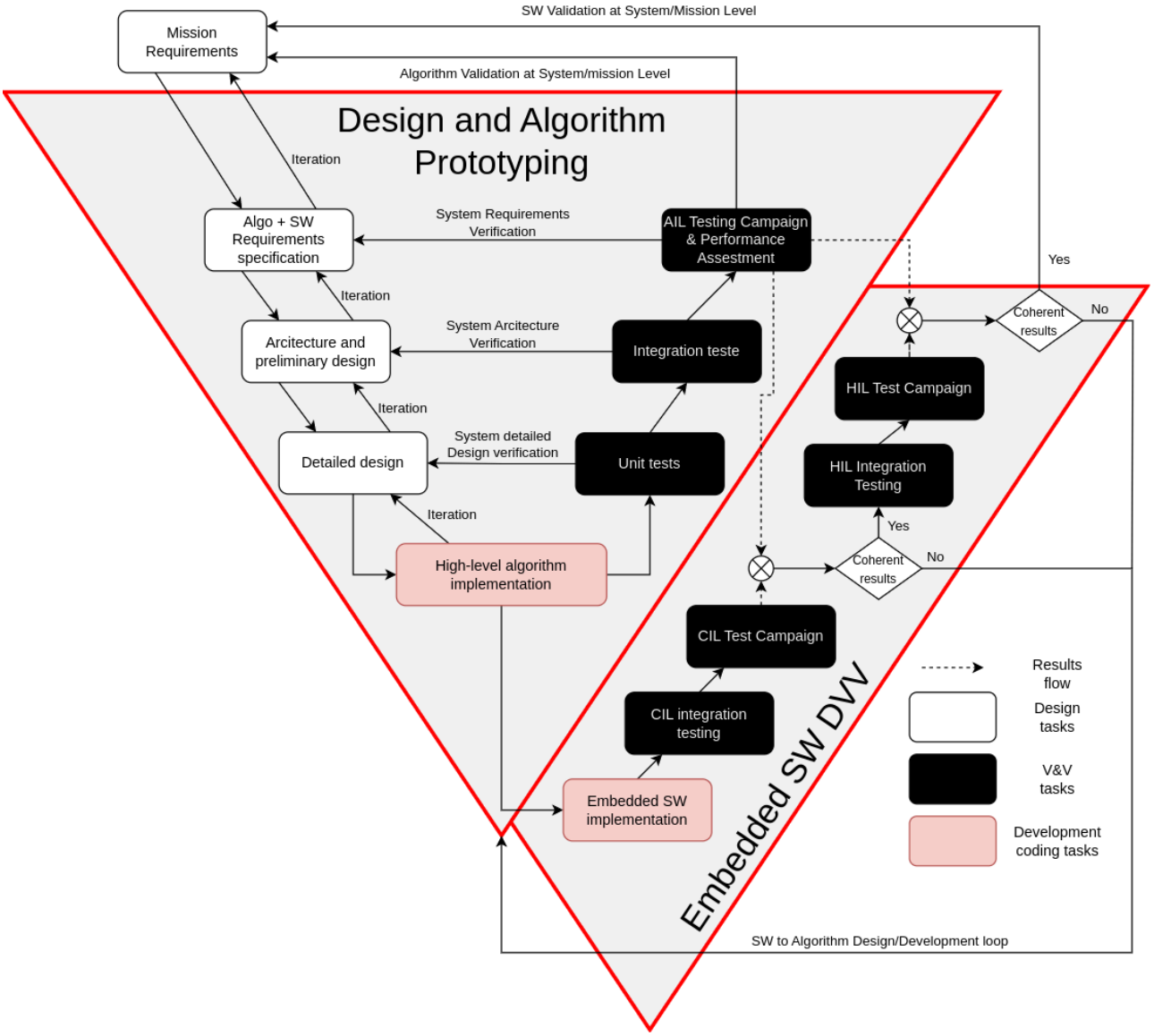

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Figure 4-1 Shows the DDVV approach for Windcube

Throughout the design and algorithm prototyping loop V-cycle, the algorithms will be designed based on the requirements specified. The detailed design will then be implemented as a prototype through a high-level programming language which will allow for rapid development and iterations though V&V tasks. An additional benefit to this approach is to rapidly test a variety of different methods for comparison.

Through the V&V cycle each subcomponent of the AOCS Prototype will be verified with unit tests, integration tests and finally the performance assessment. If the tests are not satisfactory, the design will be iterated and tests repeated.

Once the algorithm prototype has passed the V&V tasks and has been consolidated, the algorithm can be implemented as an embedded SW solution. The embedded code will go though the same V&V tasks as for the AIL. The results from the CIL test campaign will be compared with the results from the AIL test campaign to guarantee coherent results. If not, then the implementation should be revised or redesigned.

Once the CIL test campaign is passed, the integration can begin on the target hardware. The embedded implementation will then be tested through Hardware-In-the-Loop and compared with the AIL test campaign for coherent results.

FES architecture - The Functional Engineering Simulator (FES) architecture will be build with the following goals as drivers:

- Shall approach a fidelity sufficient for the confidence required by the mission scenario at hand
- Shall facilitate the Design, Development, Validation and Verification (DDVV) workflow
- Shall be flexible and scalable as required by the first goal

With these goals in mind the following architecture is proposed as depicted in Figure 4-2.

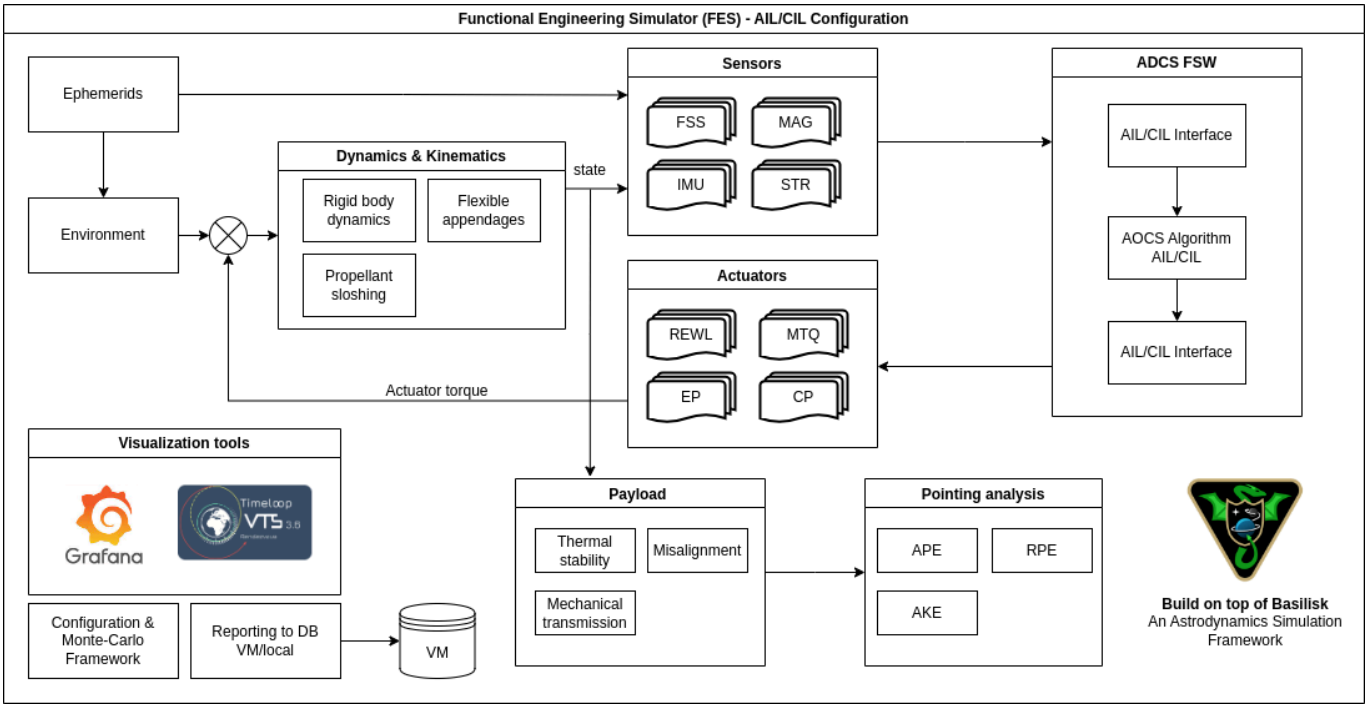

Figure 4-2 Shows the FES Architecture in a AIL/CIL Configuration

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The FES is built on top of the Open-Source Basilisk Astrodynamics simulation Framework, which is a comprehensive simulation framework written in C/C++ consisting of python modules which allows for ease of scripting and reconfigurability, while providing the execution speed of C/C++.

Basilisk already has a vast collection of dynamics, sensor and actuator models which have to be assembled in the context of the FES.

The AOCS prototype algorithms can be integrated as python modules or as embedded SW enabling the FES to run in a AIL/CIL configuration.

HIL architecture - The FES is able to be reusable for a HIL configuration running in Real-Time and providing stimuli for real hardware. This can be done by utilising the inherent flexibility of the distributed system as mentioned in the AOCS Architecture.

Each sensor and actuator is modelled as a CSP node and exposed to a CAN interface using a CAN dongle attached to the Lab PC to the HIL setup as shown in Figure 4-3.

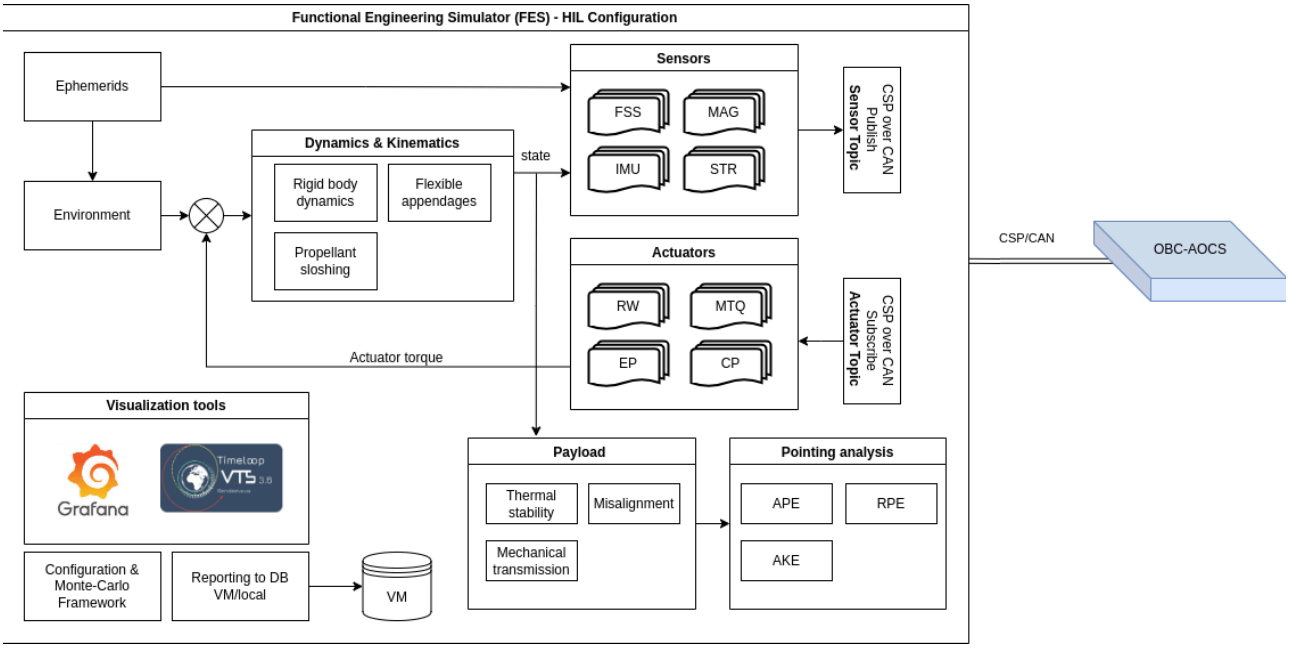

Figure 4-3 Shows the FES in a HIL configuration

For this HIL setup it is proposed to only use a reduced set of the full avionics suite. This will reduce cost and complexity. For WindCube, the HIL setup only consists of an OBC-P4 (AOCS).

The remaining avionics will be simulated and provided as stimuli to the AOCS algorithm running on the OBC-AOCS through CSP over CAN.

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5 Expected Performance

The following sections show the expected performance in the Nominal - FP mode while tracking the THRF in along- and Cross track configuration. Additionally, the slew performance will be shown as well for a slew from one THRF configuration to another.

5.1 Along-track performance

From the APE plots below it can be observed that the error slowly grows over time. This is due to the accumulated momentum in the reaction wheels. This net momentum creates a coriolis torque which will act as a disturbance. If the wheels are not dumped in time this will bring the APE outside the requirements.

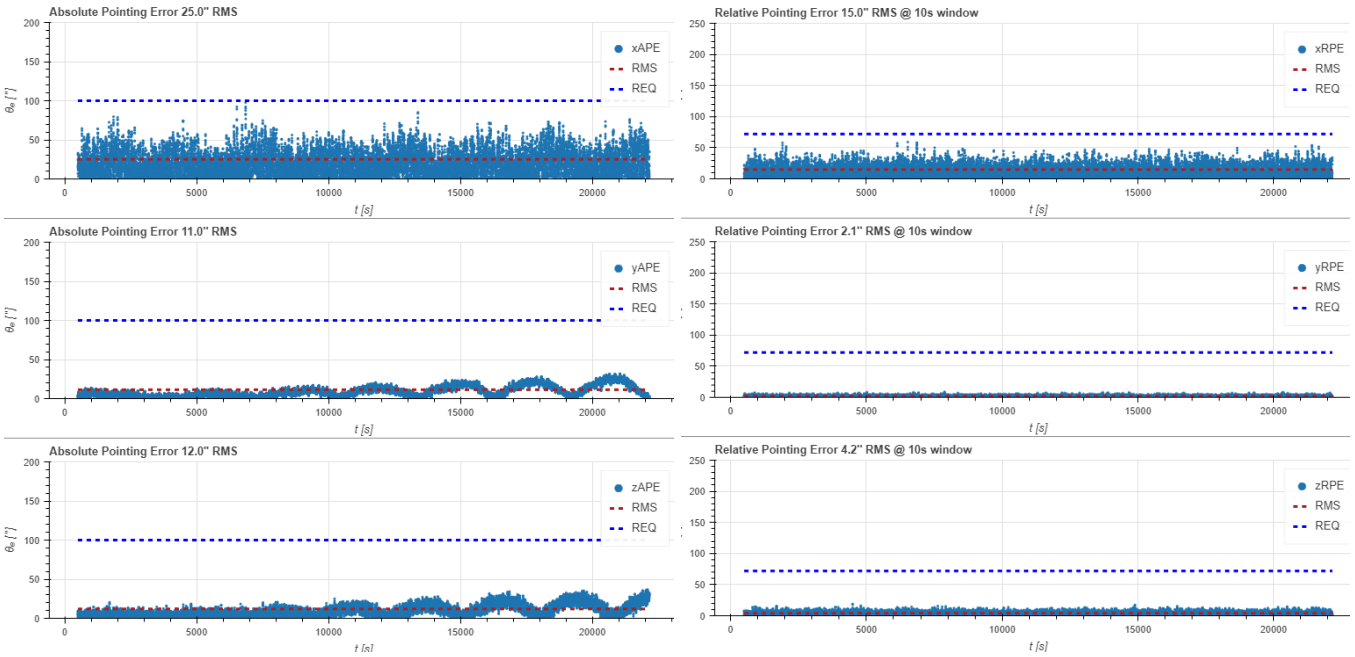

Figure 5-1: The left figure shows the APE during along-track THRF tracking, the right figure shows the RPE the same tracking reference frame.

It will be preferable to dump momentum for every time the satellite is outside the observation windows. The same applies for the cross-track results.

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5.2 Cross-track performance

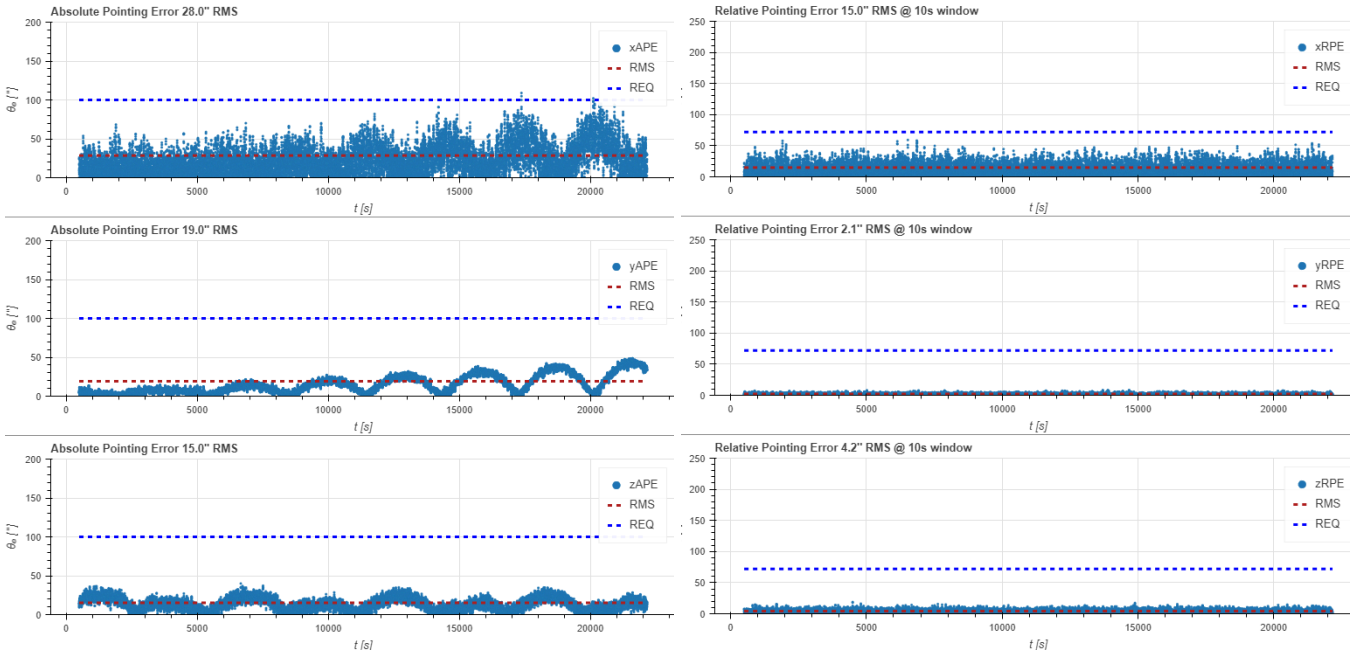

Figure 5-2: The left figure shows the APE during cross-track THRF tracking, the right figure shows the RPE the same tracking reference frame.

5.3 Attitude knowledge performance

The AKE is directly related to the star tracker performance. It can be observed that the higher noise on the boresight of the star tracker is translated to an absolute knowledge error in the x-axis.

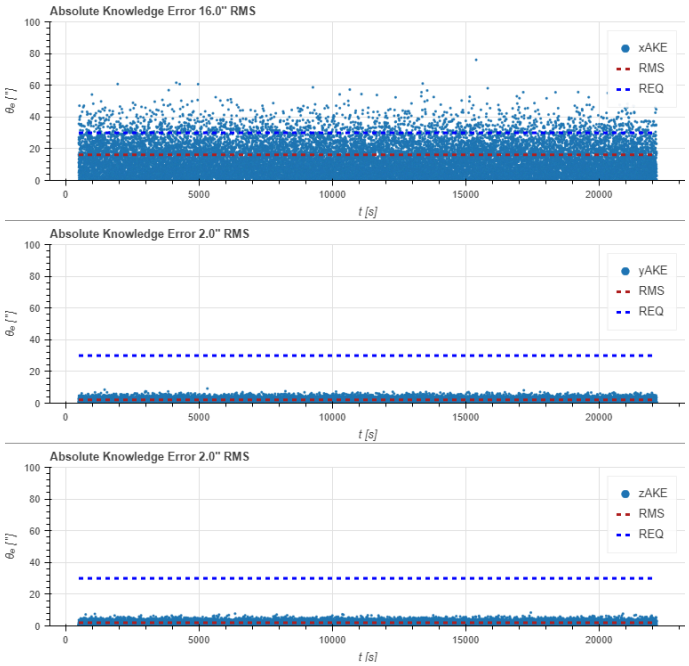

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Figure 5-3: Shows the AKE of the ADCS utilising the Star Tracker in FP mode.

5.4 THRF Guidance performance

The THRF guidance error is simulated to show the model error. The error can be kept low as long as the ADCS gets updated knowledge of the ECEF to ECI correction and the GNSS position estimate is provided.

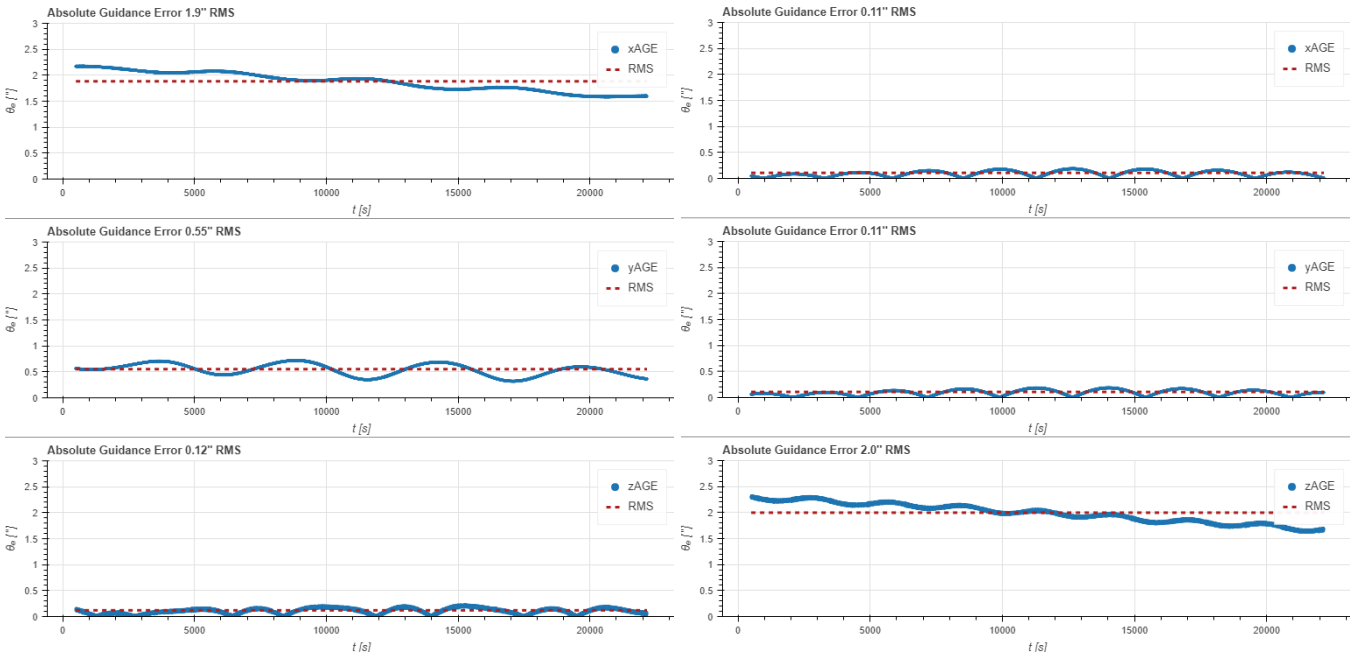

Figure 5-4: The Left figure shows the attitude guidance error for along-track THRF Configuration. Right figure shows the same for cross-track THRF Configuration.

5.5 Slew performance

The slew scenario is started from an along-track THRF configuration and switched over to cross-track THRF starting at t=0. This results in an approximate 90 deg rotation and resembles the normal use case of switching between the THRF configurations.

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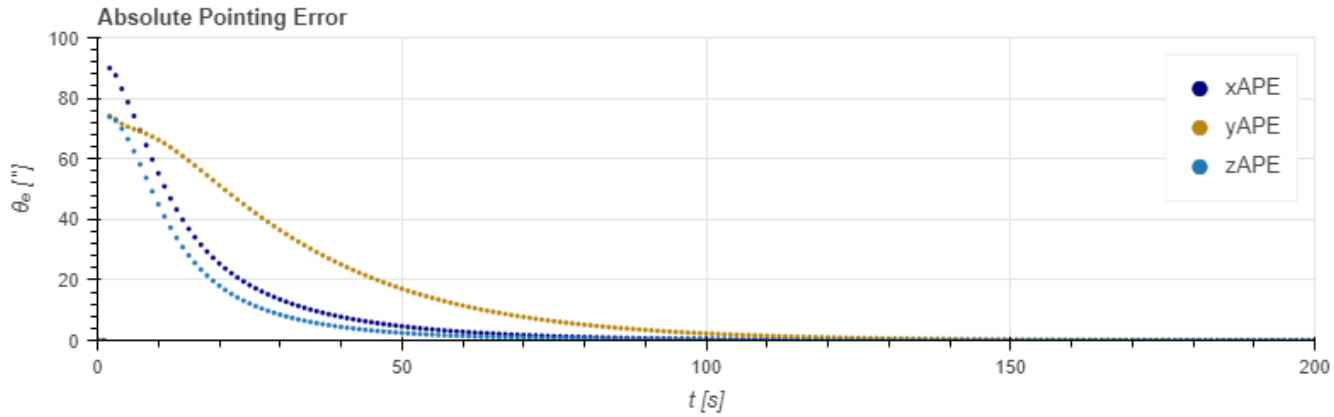

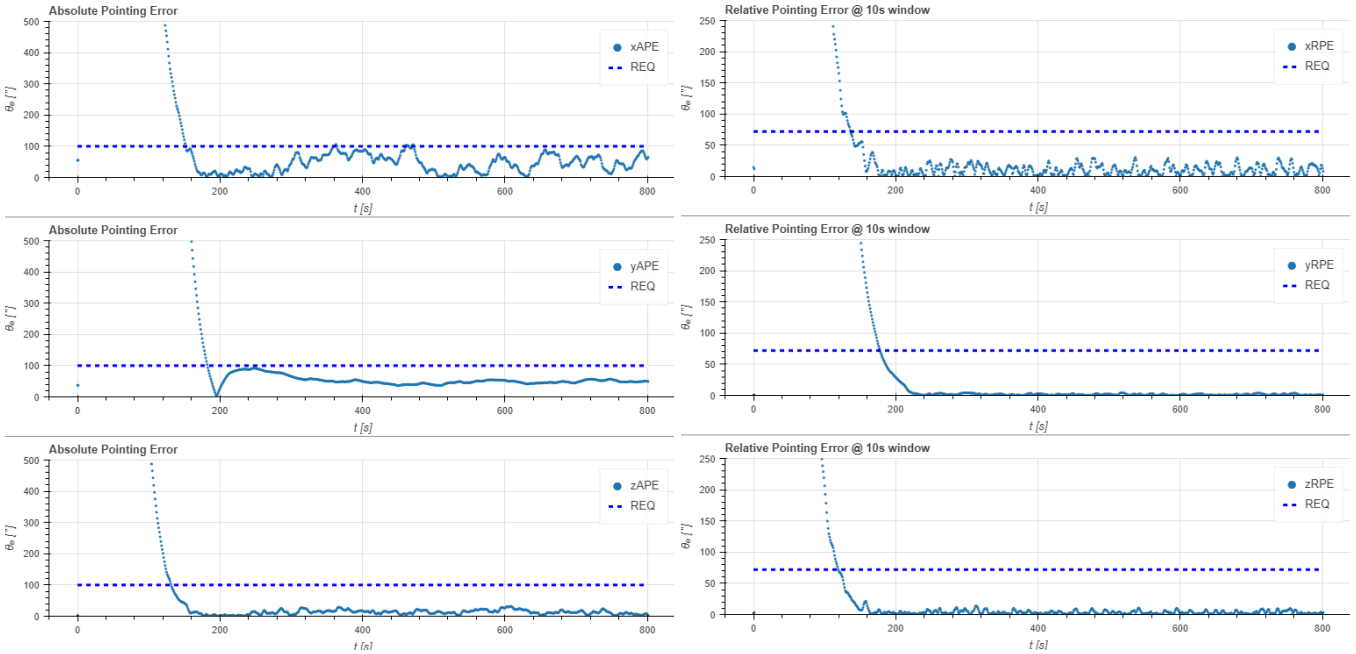

The results show that the ADCS is able to bring the APE and RPE under the requirements within the required 3 minutes.

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6 Conclusion

In summary, the analysis has been based on the driving system requirements for the attitude control system i.e:

- APE < 100 arcsec (RMS) in all axes
- AKE < 30 arcsec (RMS) in all axes
- RPE < 72 arcsec (RMS) over an integration time of 10 sec during payload operations
- The agility requirement for Windcube demands a Rest-to-Rest manoeuvre of 90 deg within 3 minutes including stabilisation to adhere to the APE requirement.
- WindCube shall track the THRF with shall have an absolute model error less than 5 km relative to the true tangent height
- The AOCS shall have the following five modes:
    - Sun pointing mode
    - Communication mode
    - Cross-track observation mode
    - Along-track observation mode
    - Deorbit mode

The analysis has shown that the ADCS is compliant with the APE, RPE, AKE and slew requirements under the assumption that the following is true:

- The solar panel cantilever frequency at 2.85 Hz is dampened to a point where the half-time is less than 10 sec.
- The ADCS attitude control bandwidth is below 0.2 Hz.
- The magnetic dipole moment is less than 88 mAm2 or the peak disturbance torque is less than 4 μNm

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